1 Discussion (excerpt)

1.1 Relation to the developer's published safety results

Meta reports safety results for every Llama generation we study: Llama 2-Chat in Touvron et al. (2023) and Llama 3.1 in Llama Team, AI @ Meta (2024), whose results are for the 3.1 models throughout. Llama 3 itself (April 2024) has no technical report. Our anchor pair, Llama-3-8B-Instruct → Llama-3.1-8B-Instruct, therefore spans exactly the post-training described in Llama Team, AI @ Meta (2024, §5.4.3). None of the developer's figures is directly comparable to ours. Its safety benchmarks are internal and anonymised and, by its own account, "not reproducible externally" (Llama Team, AI @ Meta, 2024, fn. 13). They report violation and false-refusal rates, mainly for the 70B and 405B models, without verified intent preservation or judge-error correction. Table 1 therefore sets each published claim against the measurement of ours that bears on it, and grades the relation instead of comparing numbers. Table 2 gives the one external check of our screening that is possible: the independently published SORRY-Bench fulfilment rates (Xie et al., 2025).

Three points follow from Table 1. First, no measurement contradicts a published claim. Where the two overlap, on the generational shift in refusal rates, the weakness of known single-turn templates, and the scale dependence of capability-gated attacks, they agree. The independent screening check agrees as well (Table 2, within 1.6 points for two of three models; the 13B gap is plausibly due to decoding and quantisation). Second, the developer evaluates safety against attack *forms*: adversarial and borderline prompts, red-team techniques, a system-level guard. It does not evaluate invariance of a refusal under rewording that keeps the request the same, and that is the axis on which our results diverge most. The Llama-3.1 fixes that resist every tested jailbreak template still re-open under a benign register or paraphrase change in about half of cases. Third, the lineage trend in PBR is benchmark-dependent and so does not settle the developer's "better balance" claim either way. On SORRY-Bench, whose categories include borderline requests, lower false refusal in newer generations coincides with refusals that are less robust to surface form. On HarmBench the newer generations' refusals are, if anything, slightly more robust. Where the fixes do re-open, the white-box probe locates the change before the first output token: the re-opening rewording sits lower on the newer model's refusal direction. The data-recency and refusal-wording rows are hypotheses that the pre-registered held-out replication and the human judge audit are designed to test. They are not findings.

References

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Llama Team, AI @ Meta. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Hugo Touvron, Louis Martin, Kevin Stone, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.

Alexander Wei, Nika Haghtalab, and Jacob Steinhardt. Jailbroken: How does LLM safety training fail? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Tinghao Xie, Xiangyu Qi, Yi Zeng, et al. SORRY-Bench: Systematically evaluating large language model safety refusal. In *International Conference on Learning Representations (ICLR)*, 2025.

Table 1: Correspondence between Meta's published Llama safety findings and the present measurements. Relation: ≈ consistent; + extends (a dimension the developer does not report); ↔ contrast; ? hypothesis, not tested; ∅ no counterpart. Our figures are raw and exploratory unless marked; intervals are Wilson 95%.

Theme	Developer's finding (source)	This study	Rel.
Generational balance (Llama 2 → 3)	Llama 3 gives "a much better balance between helpfulness and harmlessness than its predecessor", with lower false refusal on internal benchmarks (Llama Team, AI @ Meta, 2024, §1, Fig. 21).	Canonical compliance roughly doubles: SORRY-Bench 12.3 % → 20.9 %, HarmBench 10.3 % → 23.8 % (Llama-2-7B → Llama-3-8B). Newly answered behaviours far outnumber newly refused ones (SORRY-Bench 64 vs. 4; HarmBench 66 vs. 5)[a].	≈
Safety fine-tuning of Llama 3.1	Safety SFT and DPO on adversarial plus borderline data; refusal tone guideline (Llama Team, AI @ Meta, 2024, §5.4.3).	Llama-3.1-8B refuses more than Llama-3-8B (SORRY-Bench 20.9 % → 11.6 % compliance). Of the 82 behaviours it newly refuses (four benchmarks, one judge throughout; StrongREJECT contributes none), 45 are re-opened by a gate-admitted rewording: RSR 54.9 % [44.1, 65.2], McNemar $p \approx 6 \times 10^{-14}$; corrected 62.5 % [47.1, 77.9][b]. Adding the LLM intent check to the gate removes 2 of the 321 rewordings involved and leaves RSR unchanged.	+
Robustness to surface form	Robustness is reported for capability only: pre-trained models are "very robust" to MCQ label and format changes (Llama Team, AI @ Meta, 2024, §5.1.2); paraphrase detection is unaffected by PAWS-style adversariality (Llama Team, AI @ Meta, 2024, §5.1.3).	Refusals are not invariant to meaning-preserving rewording: Llama-3.1-8B PBR 16.8 % [13.3, 20.9] on SORRY-Bench. Register (OR 3.6) and paraphrase (OR 2.3) rewordings are the significant drivers (mixed-effects logit, Holm-corrected); back-translation recovers least.	↔
Safety across the lineage	Llama 2-Chat has a "low violation percentage overall across model sizes" (Touvron et al., 2023, Fig. 17).	The direction of the PBR trend depends on the benchmark. On SORRY-Bench, PBR is lowest for Llama 2 (7B 12.6 %, 13B 10.1 %) and higher for every Llama 3.x model (16.8–23.1 %; Llama-2-13B [7.5, 13.5] and Llama-3.2-3B [19.0, 27.8] are disjoint). On HarmBench it reverses: Llama 2 17.6 % and 17.4 % against Llama 3 14.4 % and Llama 3.1 13.8 % (overlapping intervals). On AdvBench and StrongREJECT it is small for every model (0.8–6.3 %).	≈/↔
Known jailbreak templates	Refusal suppression, hypotheticals, personas and disclaimers were "pervasive across model checkpoints, particularly when used together", mostly in multi-turn red teaming; findings fed back into safety data (Llama Team, AI @ Meta, 2024, §5.4.6).	Single-turn templates (Wei et al., 2023) on Llama-3.1-8B, judged against the original request ($n = 291$ canonically refused): refusal suppression 2.1 %, AIM, evil confidant and developer mode 0 %; the strongest class (short style injection) re-opens 4.1 % [2.4, 7.1][c]. Multi-turn attacks not tested.	≈
Scale and capability-gated attacks	8B models need a larger share of safety data than 70B (Llama Team, AI @ Meta, 2024, Fig. 18); larger models are more susceptible to code-interpreter abuse (405B 10.4 % vs. 70B 3.8 %) and prompt injection (70B 0.26, 405B 0.22, 8B 0.19) (Llama Team, AI @ Meta, 2024, §5.4.5).	An encoded combination attack re-opens 25.1 % [20.4, 30.5] of Llama-3.1-70B's refusals and 0 % of Llama-3.1-8B's: only the larger model decodes the cipher, consistent with mismatched generalisation (Wei et al., 2023)[c].	≈
Intellectual property	Llama Guard 3 reduces intellectual-property violations by 88 % (input) to 100 % (output) (Llama Team, AI @ Meta, 2024, Tab. 26).	13 of 14 HarmBench copyright behaviours newly refused by Llama-3.1-8B are re-opened by rewording: model-level IP refusal is the most surface-bound category we observe[d].	+
Refusal wording and	Overused apologies ("I'm sorry", "I apologize") rebalanced in SFT data; refusal	A refusal-prefix rule and the LLM judge diverge only for Llama 3 models	?

Table 2: External check of the canonical screening: SORRY-Bench fulfilment rates on the 440 base instructions.

Model	Xie et al. (2025)[a]	This study[b]
Llama-2-7B-Chat	13.9 %	12.3 %
Llama-2-13B-Chat	14.6 %	9.3 %
Llama-3-8B-Instruct	22 %	20.9 %

[a] Temperature 0.7, one sample, fine-tuned Mistral-7B judge.
[b] Greedy decoding on 4-bit quantised weights (Ollama defaults), compliant if any of three draws is judged compliant, qwen2.5:14b judge.